

Knowledge Bank Calculators

Full API contract — headers · params · bodies · responses

NL-FE-KB-CALC-001 v2.0 · 13 September 2026

Base: <https://nexus-lexis-backend-ql8w.vercel.app/api/v2>

0. Hard rule + headers

Snapshot replace only: PUT 11 items → store 11. No merge-by-id. Atomic validate-then-swap.

Header	On	Value
Authorization	Admin	Bearer <JWT>
X-Client-Role	Admin	Admin
Content-Type	PUT/PATCH	application/json
Content-Type	POST resource	multipart/form-data

```
Error: { "success": false, "error": "...", "message": "...", "fields": { ... } }
400 / 401 / 403 / 404 / 413 / 415 / 422
```

1. Seeded ids (8)

id	Groups
court	civil adValorem · criminal fixed · documents/family itemList
wht	purchase / sale whtBands
stamp	instruments / deeds / agri / other infoList
tax	salarySlabs slabList
business	businessSlabs slabList + surcharge fields
inherit	note
limit	causes limitList
secp	incorp incorpFee · services serviceList

GET /knowledge-bank/calculators

Auth: Public — no auth
Live tools only. Cache-Control: no-store.

Response 200

```
{
  "success": true,
  "data": {
    "schedules": [{
      "id": "court", "title": "Court fee", "status": "live",
      "hint": "...", "note": null, "resource": null,
      "groups": [{ "id": "civil", "kind": "adValorem",
        "fields": { "exemptUpto": 25000, "ratePercent": 7.5, "maxFee": 15000 },
        "items": [ ] }],
      "updatedAt": "...", "updatedBy": null
    }],
    "counts": { "live": 8, "draft": 0 }
  }
}
```

GET /knowledge-bank/calculators/:id

Auth: Public

Params

Name	Type	Req	Description
id	path	Yes	One of eight seeds

Response 200

```
{ "success": true, "data": { /* one live schedule */ } }
```

Errors: 404 if missing or draft

GET /knowledge-bank/calculators/:id/resource.pdf

Auth: Public
Raw PDF. Content-Type: application/pdf.
Errors: 404 if no uploaded file

GET /admin/knowledge-bank/calculators

Auth: Admin JWT + X-Client-Role: Admin
All eight (live + draft).

Response 200

```
{ "success": true, "data": {
  "schedules": [ /* 8 */ ],
  "counts": { "live": 7, "draft": 1 }
} }
```

GET /admin/knowledge-bank/calculators/:id

Auth: Admin

Response 200

```
{ "success": true, "data": { /* schedule incl. draft */ } }
```

PUT /admin/knowledge-bank/calculators/:id

Auth: Admin
Full snapshot. Must include ALL seeded groups. body.id must match URL.

Params

Name	Type	Req	Description
id	path	Yes	Calculator id

Request body

```
{
  "id": "court",
  "title": "Court fee",
  "subtitle": "...",
  "status": "live",
  "hint": "...",
  "note": null,
  "resource": null,
  "groups": [
    { "id": "civil", "kind": "adValorem",
      "fields": { "exemptUpto": 25000, "ratePercent": 7.5, "maxFee": 15000 },
      "items": [] },
    { "id": "criminal", "kind": "fixed", "fields": { "fee": 0 }, "items": [] },
    { "id": "documents", "kind": "itemList", "itemFeeKey": "fee",
      "fields": {},
      "items": [{ "id": "civilVakalatnama", "label": "Civil Vakalatnama", "fee": 100 } ] },
    { "id": "family", "kind": "itemList", "itemFeeKey": "fee",
      "fields": {}, "items": [] }
  ]
}
```

Response 200

```
{ "success": true, "data": {
  "id": "court", "status": "live", "groups": [...],
  "updatedAt": "...", "updatedBy": "1"
} }
```

Errors: 404 · 422 validation / missing group / kind mismatch

PUT /admin/.../calculators/:id/groups/:groupId

Auth: Admin
Table Save — replace ONE group. 10→11 stores exactly 11.

Params

Name	Type	Req	Description
id	path	Yes	e.g. limit
groupId	path	Yes	e.g. causes

Request body

```
{
  "title": "Articles & Limitation Periods",
  "kind": "limitList",
  "hint": null,
  "fields": {},
  "items": [ /* FULL list */ ]
}

// WHT: up=0 means open-ended
{ "id": "p100plus", "label": "Above 100M",
  "from": 100000000, "up": 0, "filer": 1.25, "non": 18.5 }

// Business surcharge on fields
{ "kind": "slabList",
  "fields": { "surchargeFrom": 10000000, "surchargePercent": 10 },
  "items": [{ "id": "by1", "label": "Nil", "from": 0, "up": 600000,
    "ratePercent": 0, "base": 0 } ] }
```

Response 200

```
{ "success": true, "data": { /* FULL calculator after replace */ } }
```

Errors: 404 unknown group · 422 kind/item keys

PATCH /admin/knowledge-bank/calculators/:id

Auth: Admin

Chrome only. groups/items arrays ignored.

Request body

```
{ "status": "draft",
  "hint": "Hold until rates notified.",
  "title": "Stamp duty",
  "subtitle": "...",
  "note": "..." }
```

Response 200

```
{ "success": true, "data": { /* schedule */ } }
```

POST /admin/.../calculators/:id/resource

Auth: Admin

Only tax | business. Max 10 MB PDF.

Params

Name	Type	Req	Description
id	path	Yes	tax or business

Request body

```
Content-Type: multipart/form-data

file=<pdf bytes>           // required
heading=For further information
fileName=WithholdingTaxRatesCard2027.pdf
```

Response 200

```
{ "success": true, "data": {
  "id": "tax",
  "resource": {
    "heading": "For further information",
    "fileName": "WithholdingTaxRatesCard2027.pdf",
    "url": "https://.../knowledge-bank/calculators/tax/resource.pdf"
  }
} }
```

Errors: 400 missing file · 413 · 415 · 422 wrong id

DELETE /admin/.../calculators/:id/resource

Auth: Admin — no body

Clears PDF; resource.url becomes "". FE uses bundled card.

Response 200

```
{ "success": true, "data": { "resource": { "url": "", ... } } }
```

2. Kind → required shapes

kind	fields	items[]
adValorem	exemptUpto, ratePercent, maxFee	—
fixed	fee	—
itemList	—	id, label, fee
whtBands	—	id, label, from, up, filer, non
infoList	—	id, article, label, rate
slabList	optional surcharge*	id, label, from, up, ratePercent, base
limitList	—	id, category, art, label, period, years, months, days, startFrom
incorpFee	lotSize, first/extra Online/Offline	—
serviceList	—	id, label, govt
note	—	—

3. FE swap

localStorage	API
load (admin)	GET /admin/knowledge-bank/calculators
load (public)	GET /knowledge-bank/calculators
Save tool	PUT .../calculators/:id
Save table	PUT .../calculators/:id/groups/:groupId

localStorage	API
Toggle draft	PATCH .../id { status }
PDF card	resource.url from GET